C API libhdfs

Overview

libhdfs is a JNI based C API for Hadoop's Distributed File System (HDFS). It provides C APIs to a subset of the HDFS APIs to manipulate HDFS files and the filesystem. libhdfs is part of the Hadoop distribution and comes pre-compiled in `$HADOOP_HDFS_HOME/lib/native/libhdfs.so` . libhdfs is compatible with Windows and can be built on Windows by running `mvn compile` within the `hadoop-hdfs-project/hadoop-hdfs` directory of the source tree.

The APIs

The libhdfs APIs are a subset of the Hadoop FileSystem APIs.

The header file for libhdfs describes each API in detail and is available in `$HADOOP_HDFS_HOME/include/hdfs.h`.

A Sample Program

```
#include "hdfs.h"

int main(int argc, char **argv) {

    hdfsFS fs = hdfsConnect("default", 0);
    const char* writePath = "/tmp/testfile.txt";
    hdfsFile writeFile = hdfsOpenFile(fs, writePath, O_WRONLY |O_CREAT, 0, 0, 0);
    if(!writeFile) {
          fprintf(stderr, "Failed to open %s for writing!\n", writePath);
          exit(-1);
    }
    char* buffer = "Hello, World!";
    tSize num_written_bytes = hdfsWrite(fs, writeFile, (void*)buffer, strlen(buffer)
    if (hdfsFlush(fs, writeFile)) {
           fprintf(stderr, "Failed to 'flush' %s\n", writePath);
          exit(-1);
    }
    hdfsCloseFile(fs, writeFile);
}
```

How To Link With The Library

See the CMake file for `test_libhdfs_ops.c` in the libhdfs source directory (`hadoop-hdfs-project/hadoop-`

`hdfs/src/CMakeLists.txt`) or something like: `gcc above_sample.c -I$HADOOP_HDFS_HOME/include -L$HADOOP_HDFS_HOME/lib/native -lhdfs -o above_sample`

Common Problems

The most common problem is the `CLASSPATH` is not set properly when calling a program that uses libhdfs. Make sure you set it to all the Hadoop jars needed to run Hadoop itself as well as the right configuration directory containing `hdfs-site.xml`. It is not valid to use wildcard syntax for specifying multiple jars. It may be useful to run `hadoop classpath --glob` or `hadoop classpath --jar <path>` to generate the correct classpath for your deployment. See Hadoop Commands Reference for more information on this command.

Thread Safe

libdhfs is thread safe.

- Concurrency and Hadoop FS "handles"

  The Hadoop FS implementation includes a FS handle cache which caches based on the URI of the namenode along with the user connecting. So, all calls to `hdfsConnect` will return the same handle but calls to `hdfsConnectAsUser` with different users will return different handles. But, since HDFS client handles are completely thread safe, this has no bearing on concurrency.

- Concurrency and libhdfs/JNI

  The libhdfs calls to JNI should always be creating thread local storage, so (in theory), libhdfs should be as thread safe as the underlying calls to the Hadoop FS.